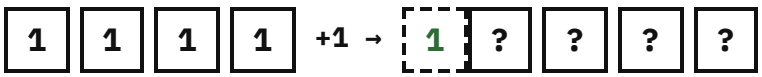
UNIT 1 · LESSON 5 · CASE 1

The Broken Odometer

Team: _____

Roles (switch each case): **Decoder** works the bits · **Checker** verifies every step · **Runner** brings the checkpoint to the teacher.



A tiny computer stores numbers in **exactly 4 bits**. Like a car odometer, it can't add a new digit when it runs out of room.

Vocabulary

overflow error	round-off error

Part A: Whole numbers place values 8, 4, 2, 1

1. What is the largest number 4 bits can store?

☐ ☐ ☐ ☐ = _____ in decimal

2. The computer holds 1111 and adds 1. What does it hold now?

☐ ☐ ☐ ☐ Error name: _____

Part B: Engineers relabel the bits place values 2, 1, ½, ¼

Place value	2	1	½	¼	Value
3. Store 2.75					2.75
4. Store 1.3 (closest you can)					
5. Largest value possible					

4b. For #4: how far off is your answer from 1.3? _____ Error name: _____

5b. For #5: what happens if you add 0.5 to the largest value? Why?

_____

_____

CHECKPOINT 1 Bring your answers to **#2** and **#4**.

teacher

UNIT 1 · LESSON 6 · CASE 2

The Secret Message

Team: _____

H → 72 → 01001000 Every character you type is stored as a **number**, and every number is stored as **bits**. Sender and receiver just have to agree on the code.

Part A: ASCII each capital letter is one 8-bit byte

A	B	C	D	E	F	G	H	I	J	K	L	M
65	66	67	68	69	70	71	72	73	74	75	76	77

N	O	P	Q	R	S	T	U	V	W	X	Y	Z
78	79	80	81	82	83	84	85	86	87	88	89	90

Place value	128	64	32	16	8	4	2	1

1. Decode the message.

01000010	01011001	01010100	01000101
# _____ letter _____	# _____ letter _____	# _____ letter _____	# _____ letter _____

Part B: Class code 5 bits each · space = 0, A = 1, B = 2, … Z = 26

2. Decode the message.

00011	01111	00100	00101
# _____ letter _____	# _____ letter _____	# _____ letter _____	# _____ letter _____

3. Encode your first name in the class code.

_____

4. Why are 5 bits enough for 27 symbols?

_____

5. Add lowercase letters too (53 symbols). How many bits per symbol now? _____

6. Why can't 8-bit ASCII handle every language and emoji? What does the world use instead?

_____

CHECKPOINT 2 Bring **both decoded words**.

teacher

UNIT 1 · LESSON 7 · CASE 3

The Hidden Picture

Team: _____

The rules. The first 8 bits are the image's **width**. The next 8 bits are its **height**. Every bit after that is one pixel, row by row, left to right. **1** = black **0** = white

The file

`00001000` `00001000` `01100110` `11111111` `11111111` `11111111` `01111110` `00111100` `00011000` `00000000`

Green bytes are the header (metadata), not pixels.

	1	2	3	4	5	6	7	8
1								
2								
3								
4								
5								
6								
7								
8								

1. Width = _____
Height = _____

2. Shade the grid. What is the picture?

3. How many bits is the whole file, header included?

4. The same picture at 16 × 16 pixels: how many pixel bits?

5. A real sunset is **analog**. Why is any digital image only an approximation of it?

CHECKPOINT 3 Bring **the picture** and your answer to **#3**.

teacher

UNIT 1 · LESSON 8 · CASE 4

The Color Lab

Team: _____

R G B Each pixel uses **3 bits**: one for **R**ed, one for **G**reen, one for **B**lue light. A 1 turns that light on. Mixing light makes the other colors.

Color key

000 black	**100** red	**010** green	**001** blue
110 yellow	**101** magenta	**011** cyan	**111** white

The file 4 pixels wide, 2 pixels tall, row by row

`100` `110` `010` `001` `111` `101` `011` `000`

1. Color the grid (or write each color's name).

	1	2	3	4
1	color:	color:	color:	color:
2	color:	color:	color:	color:

2. How many colors are possible with 3 bits per pixel? _____

3. How many colors with 2 bits per color (6 bits per pixel)? _____
With 8 bits per color (24 bits per pixel)? _____

4. A 1920 × 1080 photo uses 24 bits per pixel. How many **bytes** is it, before compression? Show your work.

5. Why might an app choose fewer bits per pixel? What does it give up?

_____

CHECKPOINT 4 Bring the colors in **row 2** and your answer to **#4**.

teacher